Michelin Group
& Intelligent Tires

Three Main Types of Potential Utilizations for Electronics

1. Identification
2. Operating Conditions
3. Vehicle Performances

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Michelin Leadership Mission Statement

- **We favor an RfID “single standard” solution including migration to EPC codification.**
  - We have worked to condense these standards into a “single standard”. ISO 18000-6c (EPC Gen 2) as the B11 tire standard achieves that.
- **We are developing RfID capabilities both as a supply chain tool and as added value to our products.**

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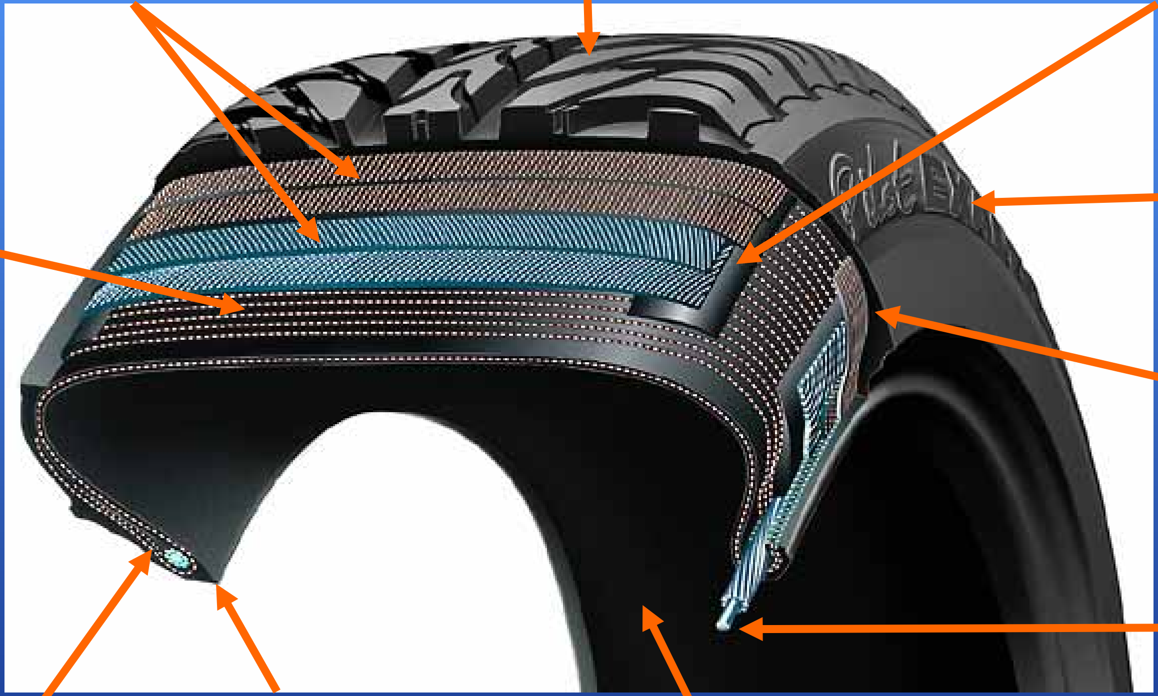
The structure



Home

Detuning by Carbon in Rubber

Imbedded RFID place in:	Imbedded Read Distance (patch on inner surface)	
	0.6-1.2 meters	(1.0 +)
passenger tire	0.4-0.8 meters	(about 1.0)
aircraft	0.2-0.4 meters	(0.6-1.0)
truck	0.1-0.2 meters	(0.2-0.3)
earth mover		

Integration - RfID Label

- Where should the tag be affixed?

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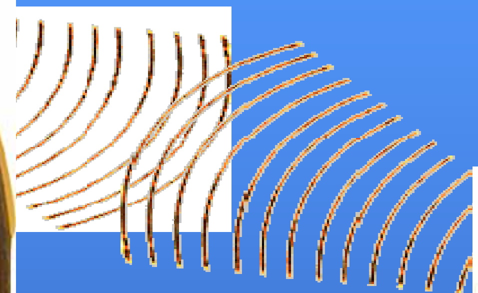
Labels, Labels, and More Labels!

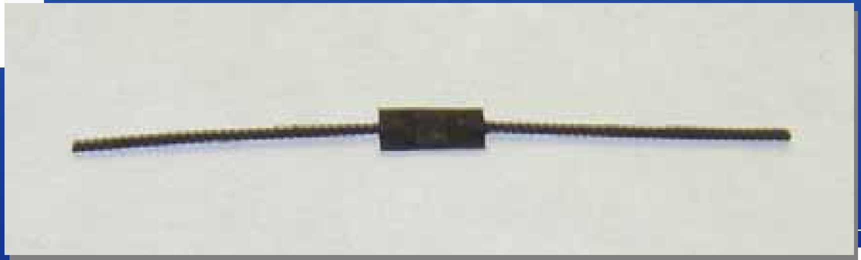
If you have this:

You might not need this:

Or these:

TODAY MICHELIN APPLIES UP TO 12 OPTICAL MARKS FOR OE's





Licensed Global Electronics Suppliers to insure availability (4 currently)

- **Licensee's:**
- **AWID - Part Number # APT-2024**
- **Sharp Electronics (when Japan market starts)**
- **ACG – recent licensed**
- **HanaRFID**
  - HanaRFID has recently initiated manufacture and distribution part number HTAG0001 ($0.89)
  - HanaRFID recently announced relationship with AdvancedID for $0.40 (10 million volume)

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Advanced ID Establishes Pricing Breakthrough & Offers US$0.40 UHF - RFID Tire Tags to the Global Tire Industry

April 20, 2006, Advanced ID Corporation (OTCBB: AIDO) Vancouver WA - Advanced ID Corporation in cooperation with major global tire manufacturers is responding to the industry needs for high-quality and cost-effective RFID tire tags. The 40 cents per tag price goes into effect once a total quantity of 10 million tags is ordered to activate the pricing. The global tire manufacturing industry has an annual automotive and light truck production level of approximately 700 million tires.

Licensed Global Electronics Suppliers to insure availabilty

- Permanent patch provided by Patch Rubber Inc.
    - Part Number # 98270

Patch RfID

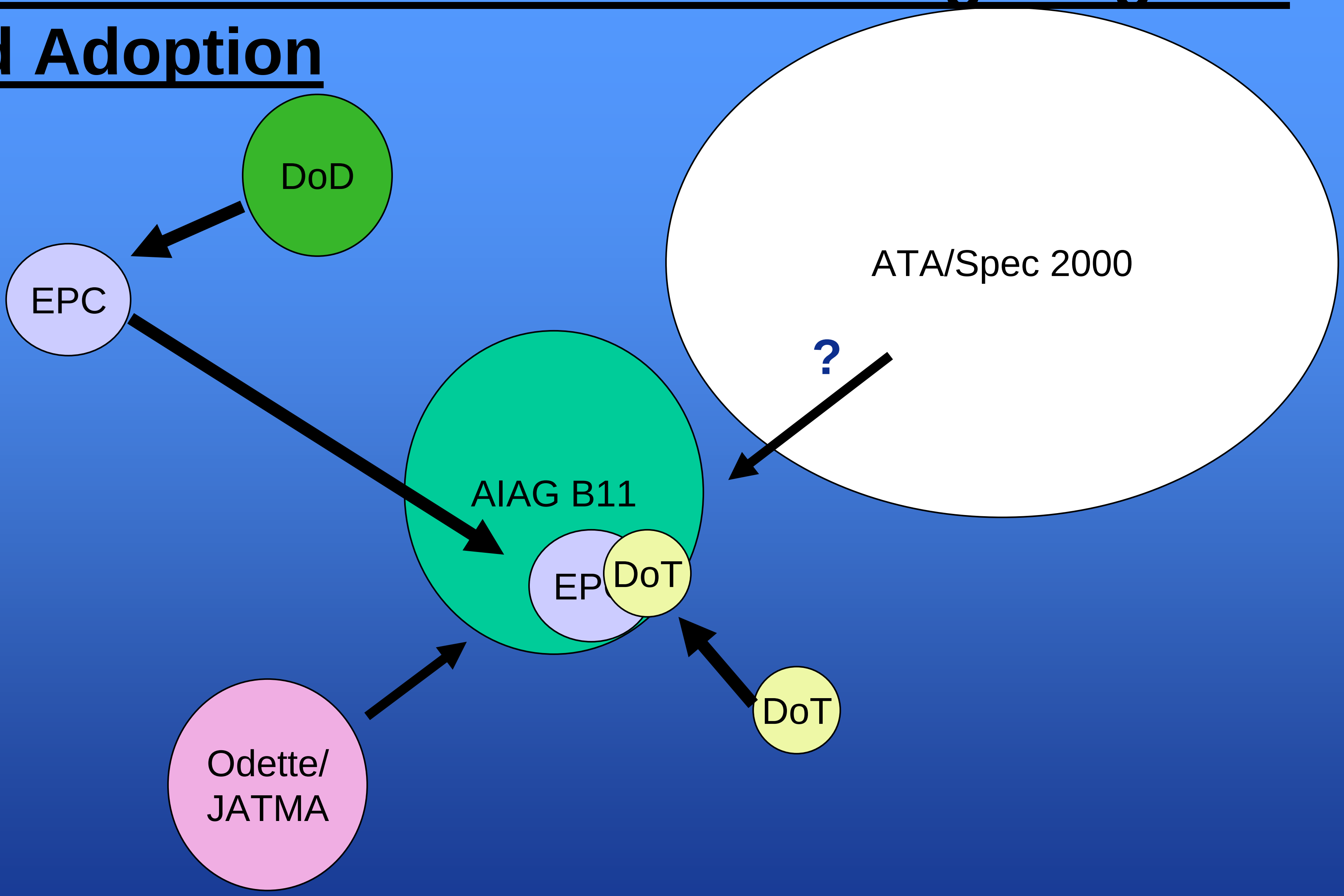
- RfID is cured in tag.

Tire Data Standard Harmonization History

AIAG B11 – History Automotive Industry Action

- **20 Feb 2001 – First AIAG Tire and Wheel Identification task force meeting was convened.**
- **15 Feb 2002- B-11 standards for RfID Logistics established**
- **6 Jan 2003 Michelin announces commercial solution.**
- **31 March 2004 <u>Michelin petitions</u> to reopen B11 and make it ePC compliant.**
- **18 May 2004 AIAG and ePC work together to complete revised B11 that complies with EPC.**
- **Expected 2005 – Migration to ISO 18000-6c**
- **<u>KEY Meeting Dec 7,2005 in Detroit (EPC, Odette,JAMA,JAPIA, AIAG, DoD, OE and Tire manuf. to create single standard = same RFID to OE and Retail and Military)</u>**
- ISO 18000 -6c and EPC Gen 2 modified to account for user data as a result of the B11 Harmonization.

2004 -2005 Data Standards Driving Progress and Adoption



Key Michelin Role: RfID Standards Linked to Tires Migrating to Single Standard

- **RfID (2000-2004)**
  - B-11 (AIAG Tire and Wheel Standard)
  - ePC (Class 0 and Class 1) (Electronic Product Code for Retail)
  - ISO-18000-6 a,b (International Standard for Logistics)

- **RfID Emerging To Single Standard (2005)**
  - ePC UHF Gen 2 / ISO 18000-6c

- **Michelin Lobbied in 2004**
  - Asia- JATMA, Korea, China Sparkice (Bridgestone, Yokohama, Toyo, Sumitoma, Sharp, Omron, Inter-Active, Taechu
  - Europe- VDA , ITA, Germany, Galia France (VW,BMW,Audi) Odette
  - US- AIAG, DoD , FAA (Boeing/Airbus) (Goodyear, Continental. Pirelli, Firestone)
  - International- ePC, UCC.EAN, AIM Global (REG).

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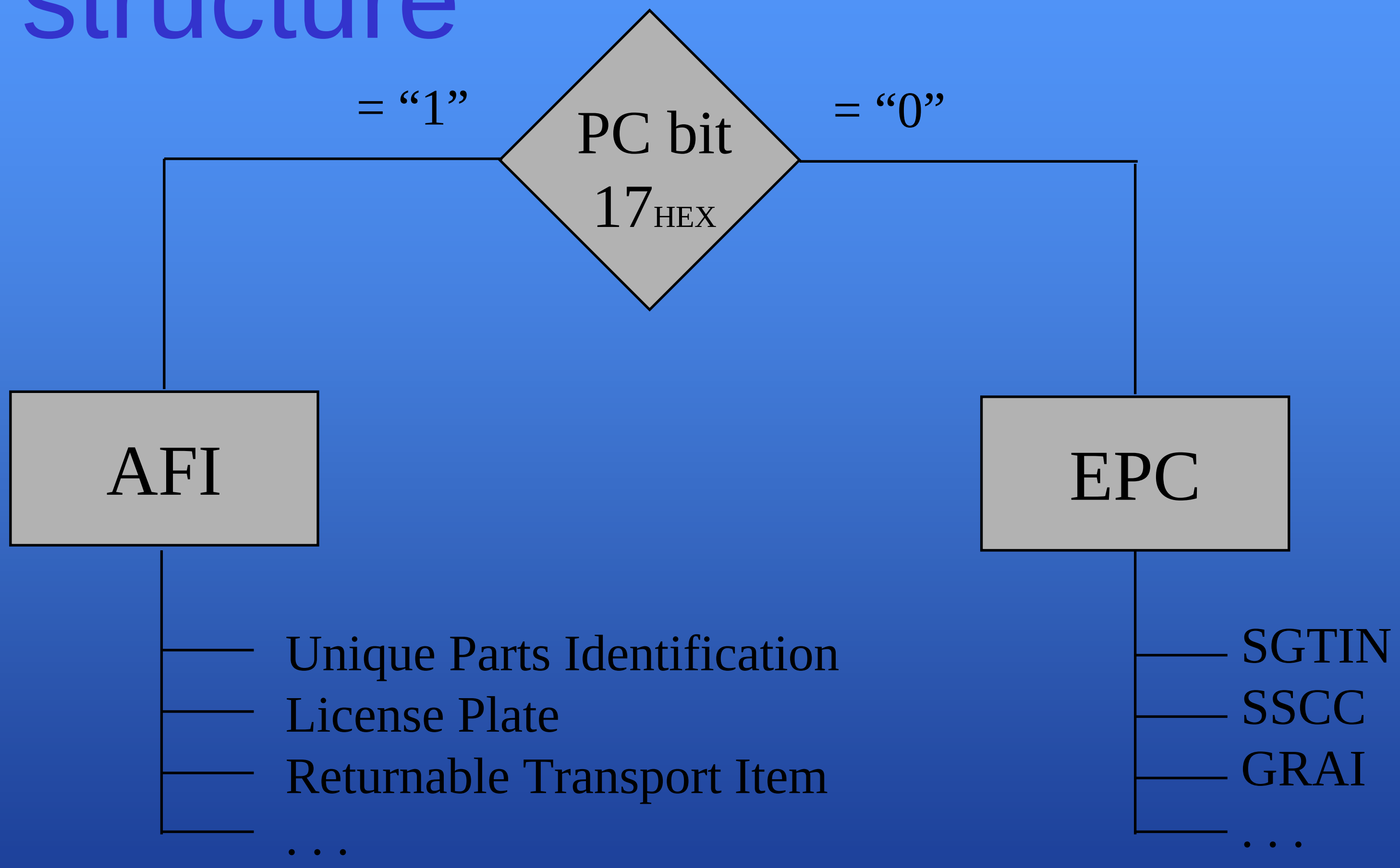
Basic RFID user information structure



Tire Serialization = Bar Code Number + Serial Number

- **Today**
- **Bar Code (UPC) Manufacturer # + Product #**

- **Evolving to:**
- **Electronic Product Code (EPC)**

  Manufacturer # + Product # + Serial # + DI

DI = Data Identifier

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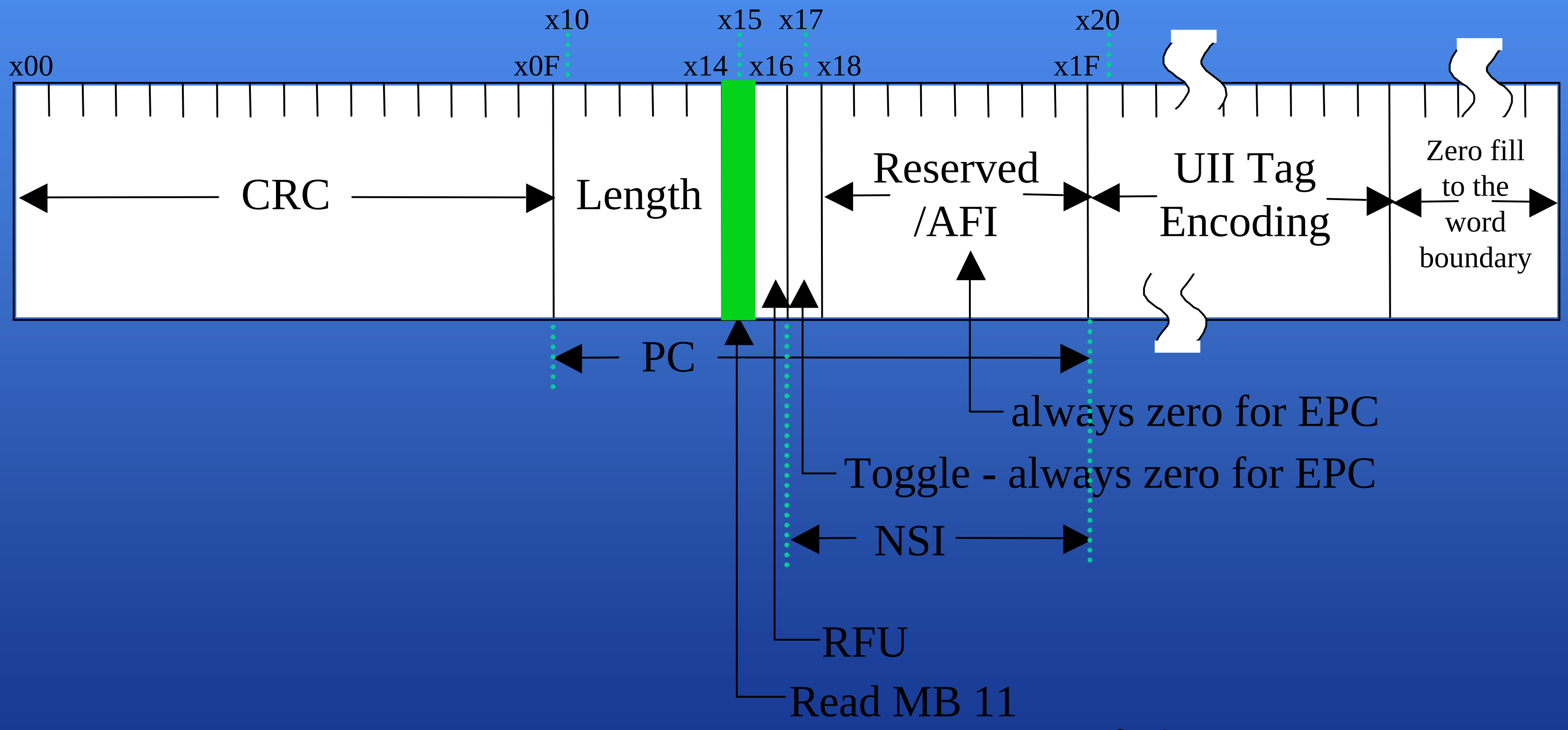
Challenge for B11 and EPCglobal

- **Multiple industries will use the same UHF air interface protocol**
- **Different sectors will use different identifier encodings**
  - Commercial, DoD, Retail – EPCglobal Tag Data Standard
  - Aero & Industrial – 15961/962, SPEC2000, other
- **How can tires be identified across multiple sectors?**

Bit 15 and Bit 17



End User examples

MB01 15 \ MB01 17	0	1
0	Yes EPC No User Data (Would be appropriate for an exclusive retail tire)	No EPC No User Data
1	Yes EPC Yes User Data (B-11 will state that all tires use this method)	No EPC Yes User Data (Would be appropriate for automotive OEM)

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RFID Physical Standards

- AIM Global TOR 5Q: Proposed Guidelines for the Verification and Qualification of Design and Manufacture for RFID Chips and Transponders For Tires
- ISO17367- Supply Chain Applications of RFID-Product tagging Annex incorporated devices.
- No Current Gen 2 chip has reuqired 1Kbit user memory for ITEM level Tire Standard (B11).
- Click here for video of tire application

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RFID in the Automotive Industry

Key Challenges

Recent Notes from Automotive Meeting in Stuttgart

B. Gregory (Ford Europe) and Dr. Roya Ulrich (DaimlerChrysler)

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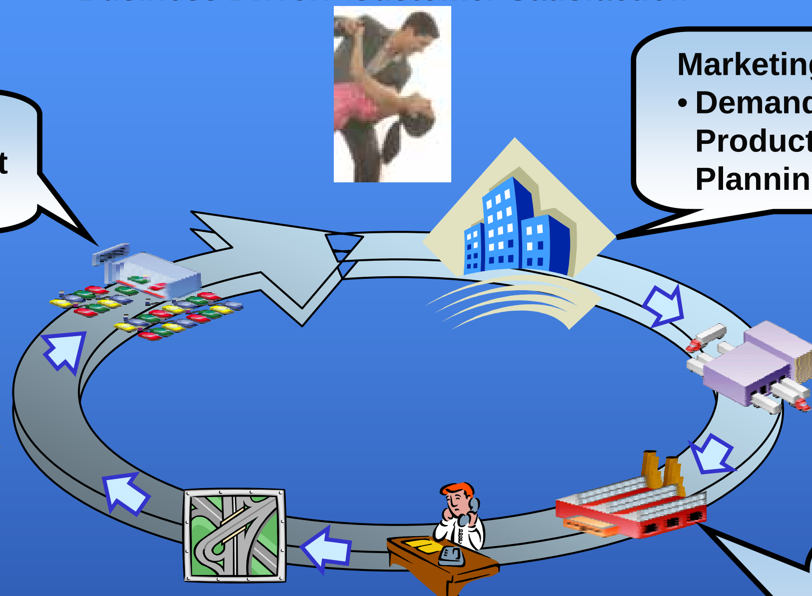
RFID Opportunities in The Automotive Value Chain →

The Automotive Industry Value Stream

Business Driver: Customer Satisfaction



Dealerships:
- Compound Mgm't
- Vehicle Locator

Marketing:
- Demand driven Production Planning

Supply Chain:
- Mat. Moveme
- Delivery Accu
- Shipment control

Service:
- Traceability for re calls and warran
- Service parts
- Parts Warehouse Mgmt

Shop Floor:
- Process Control
- KanBan
- Part Replenishment
- Mat. Movements
- WIP tracking

Traffic:
- Veh. Logistics
- Compound Mgm't
- Delivery Estimate
- Delivery Tracking

Internal Efficiencies:
- Container Tracking
- Inventory Mgm't
- Asset utilization
- Quality Control

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Key Management Challenges

- Recent advances in the development and application of RFID technology in other industries (Retail, Aerospace, DoD, etc.) is causing the prices for equipment and tags to drop, thereby reducing the cost/benefit ratio for pilot projects and increasing the return on investment. **However, because the Automotive Industry does not speak with one voice, RFID vendors and organisations that set technical and data standards are not focusing on Automotive applications and needs.**

- As every Automotive Manufacturer fights to improve their profit picture, pressure continues to build to reduce cost including major cross functional and enterprise processes Across the Automotive Value chain, e.g. Supply Chain Management, Manufacturing Material Flow, Vehicle Logistics, etc.

- Each Automotive Manufacturer up to now has begun to test the benefits of RFID applications with Proof of Concept pilots. This is leading to a fragmented deployment approach, both within Automotive OEM's own enterprise as well as in joint pilots with their key strategic partners, suppliers and service providers.

- This fragmented approach also applies to the OEMs approach to standards organisations, thereby compounding the problem.

Opportunities

- The Tire and Wheel Industry has a requirement to work with many different industries to label their products with RFID, e.g. Wal-Mart in the Retail Industry, Boeing in the Aircraft Industry, etc. For this reason they have actively promoted the development of unique RFID labels and products for Tires and a flexible data standard called “B11” to cope with different data formats in the various industry eco-systems.

- **The Automotive Industry has a unique opportunity to adopt the B11 data standard as a “normative standard” and as an enabler to bridge the gap between different data formats with different product coding (e.g. EPC, ISO AFI, ...), across the different industry sectors (e.g. Commercial, Retail, Airline, DoD, Automotive, ...)**

- This opportunity would also enable synergies and standards for other Automotive Eco-systems, such as the Container Industry, as well as for Container and Asset Management processes by the OEMs.

- Collectively these opportunities would become a Strategy for a „Fast-track“ approach to „Kick-start“ the Automotive Industry and „Leap-frog“ other „early adopters“ of RFID technology in Commercial, Retail, Airline, DoD and other industries to achieve radical business process improvements and significant ROI.

Options

<u>Objectives</u>:

- Define a **Strategy** for a “Fast-track” approach for the Automotive Industry that would consolidate and synchronise future RFID related implementaitons and standards
- Define a **Roadmap** for future processes/applications between Automotive Manufacturers and their key partners, suppliers and service providers to define common solutions and standards for the Automotive Industry
- Build on **Synergies** to avoid duplication and unnecessary workload to speed up the proposal process and increase benefits maximize ROI.

<u>Option 1</u>:

- Defines an organisational structure composed of known standards organisations at national, regional and international level under the leadership of the <u>Joint Automotive Industry (JAI)</u> and driven by a new Unified Automotive RFID Business Action Group

<u>Option 2</u>:

- Defines a Global organisational structure led by <u>EPC Global</u> and driven by a new EPC Business Action Group for Automotive

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April 19, 2006

TO: Radio Frequency ID Work Group / Automatic Identification and Data Collection Advisory Group (RFID-AIDC0411)

SUBJECT: Minutes from the April 11, 2006 Meeting

RFID OPPORTUNITIES IN THE AUTO VALUE CHAIN

- The Automotive Industry has a unique opportunity to adopt the B-11 Tire and Wheel Identification Standard as a 'normative standard'. The B-11 would become an enabler to bridge the gap between different data formats and product coding schemas (i.e. EPC, ISO AFI, etc.) which span several industry sectors (e.g. Commercial, Retail, Airline, the DOD, and Automotive).
- The AIAG B-11 Tire and Wheel Label and Radio Frequency Identification Standard is the world's first item-level RFID tracking and traceability standard.

Summary

- A new harmonized RFID standard for tire serialization now exists.
- Michelin has the core capability to include RFID as the commercial opportunity arises.
- Commercial sectors like truck tires and aircraft tires are the most likey “early adoption arena’s” based on business values

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